

System Design

Subject Code: TCS-504

Assignment 1

Movie Ticket Booking System



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A Requirement analysis (FR + NFR)

Functional Requirements

The system shall display the list of all movies currently playing in the cinema.

The system shall display all available shows, including the screen number and show time, for the movie selected by the customer.

The system shall display the seat layout of the selected show, indicating whether each seat is AVAILABLE or BOOKED.

The system shall allow the customer to book one or more available seats. If any selected seat is already booked, the system shall reject the booking for that seat.

The system shall calculate the total booking amount based on the seat type: Silver – ₹150, Gold – ₹250, Platinum – ₹400.

The system shall allow payment through UPI, Card, or Cash. If the payment fails, the booking shall not be confirmed.

The system shall generate a ticket containing the booking ID, movie name, screen number, show time, booked seat numbers, and total amount after successful payment.

The system shall allow the customer to cancel an existing booking, after which the booked seats shall become AVAILABLE again.

Non-Functional Requirements

The system shall follow a modular design by implementing one class per file to improve maintainability.

The system shall be extensible so that new payment methods can be added with minimal modification to the existing code.

The system shall validate invalid menu choices and seat numbers and display clear error messages without crashing.

The system shall follow Object-Oriented Programming principles to produce clean, maintainable, and reusable code.

Relationship Table

Pair	Relationship	Justification (Lifetime Test)
Cinema — Screen	Composition	A screen belongs to one cinema. If the cinema is removed, its screens no longer exist.
Screen — Seat	Composition	Seats are permanently part of a screen. Destroying the screen also removes its seats.
Show — Movie	Aggregation	A movie exists independently of any particular show and can be scheduled in multiple shows.
Show — Screen	Association	A show is conducted on a screen, but neither object owns the other. A screen continues to exist even if a show is removed.
Show — ShowSeat	Composition	ShowSeat objects exist only for a specific show. When the show is deleted, its ShowSeat objects are also removed.
Booking — Customer	Association	A customer can exist without any booking, and bookings reference customers without owning them.
Booking — ShowSeat	Aggregation	A booking refers to selected ShowSeat objects but does not create or own them. ShowSeat objects continue to exist after booking cancellation.
Booking — Payment	Association	Payment is used during the booking process, but Booking does not own the Payment object permanently.
Payment — UpiPayment	Inheritance	UpiPayment is a specialized type of Payment and inherits its interface.
Payment — CardPayment	Inheritance	CardPayment extends Payment by implementing card-specific payment behaviour.
Payment — CashPayment	Inheritance	CashPayment extends Payment by implementing cash payment behaviour.
BookingService — Booking	Association	BookingService creates and manages bookings during execution but does not own their lifetime.

Noun-verb analysis

Noun Found	Keep as a Class?	Reason
Movie	Yes	Has its own data and identity.
Seat	Yes	Has seat number, type, and price.
Seat Layout	No	It is only a view of a show's seats, not an independent object.
Cinema	Yes	Represents the theatre containing multiple screens.
Screen	Yes	Represents an auditorium and owns its seats.
Show	Yes	Represents a movie screening at a particular time and screen.
ShowSeat	Yes	Represents the status of a seat for a particular show.
Customer	Yes	Represents the person booking or cancelling tickets.
Booking	Yes	Stores booking details such as seats, customer, and amount.
Payment	Yes	Represents the payment abstraction.
UPI	No	It is a specific payment method (implemented as UpiPayment), not a separate domain entity.
Card	No	It is a specific payment method (implemented as CardPayment).
Cash	No	It is a specific payment method (implemented as CashPayment).
Ticket	No	It is generated as output by TicketPrinter, not maintained as an independent entity.
Booking ID	No	It is an attribute of the Booking class.
Screen Number	No	It is an attribute of the Screen class.
Start Time	No	It is an attribute of the Show class.
Price	No	It is an attribute/value associated with seat types.
Price Calculator	Yes	Has the single responsibility of calculating the total booking amount.
Ticket Printer	Yes	Has the single responsibility of formatting and printing tickets.
Booking Service	Yes	Coordinates the complete booking process.

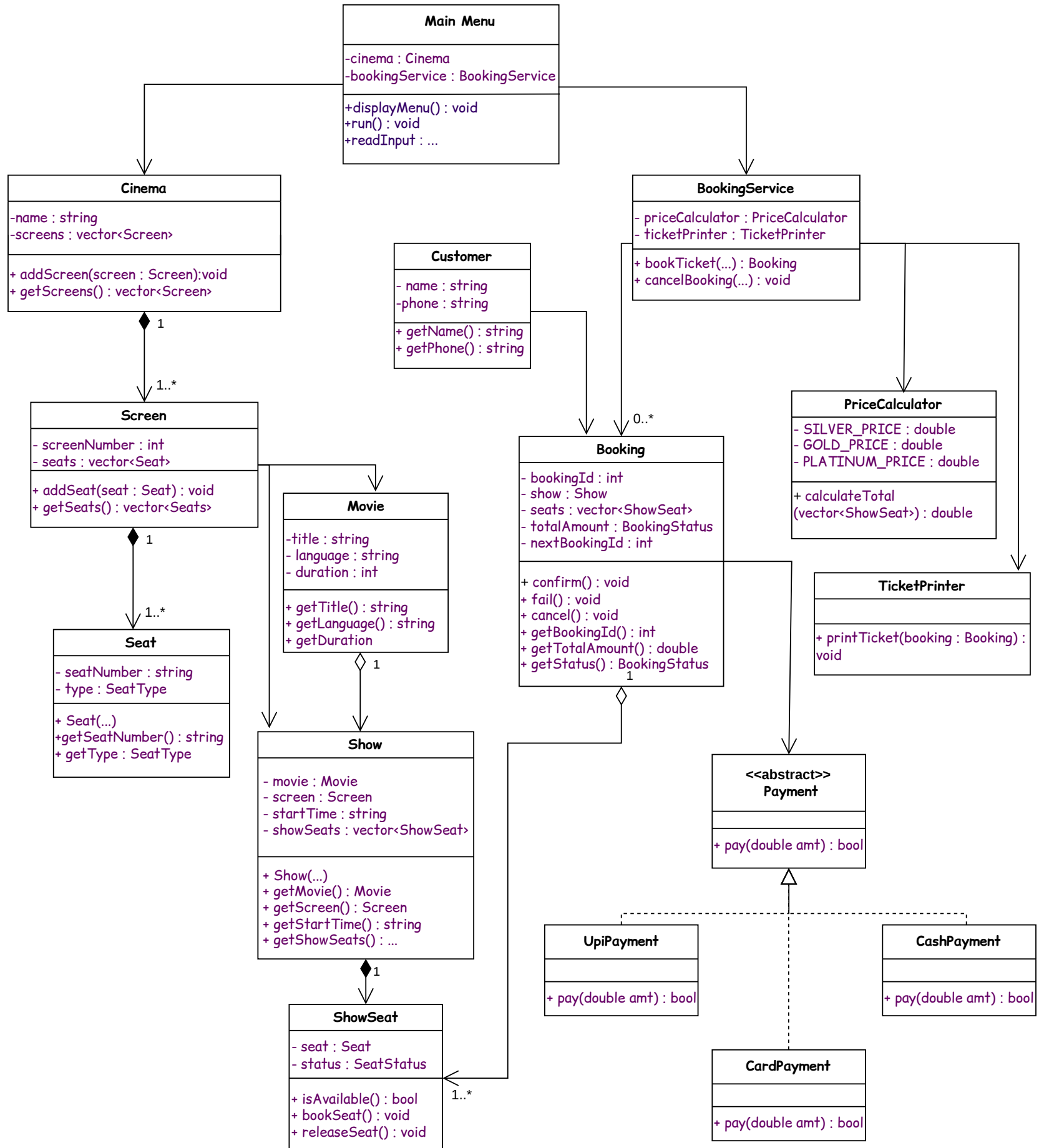
Core Entity Classes

Class	What it Knows (Attributes)	What it Does	What it Must NOT Do
Movie	Title, language, duration	Stores and provides movie information	Must not manage bookings or payments.
Seat	Seat number, seat type	Stores seat details	Must not store booking status.
Screen	Screen number, collection of seats	Maintains seats for a screen	Must not schedule shows.
Cinema	Cinema name, collection of screens	Maintains screens and available movies	Must not process bookings.
Show	Movie, screen, show time, collection of ShowSeats	Represents a movie screening and manages seat availability	Must not process payments.
ShowSeat	Seat reference, booking status	Marks seats as available or booked	Must not calculate ticket price.
Customer	Name, phone number	Stores customer information	Must not confirm bookings.
Booking	Booking ID, show, selected seats, total amount, booking status	Stores booking details and booking status	Must not print tickets or perform payments.

Behaviour / Service Classes

Class	What it Knows	What it Does	What it Must NOT Do
Payment (Abstract)	Payment amount	Defines the payment contract pay(amount).	Must not implement a specific payment method.
UpiPayment	UPI payment details	Processes payment using UPI.	Must not manage bookings.
CardPayment	Card payment details	Processes payment using Card.	Must not calculate prices.
CashPayment	Cash payment details	Processes payment using Cash.	Must not manage booking data.
PriceCalculator	Seat prices	Calculates the total booking amount.	Must not store booking information.
TicketPrinter	Booking details	Formats and prints the ticket.	Must not process payments.
BookingService	References to booking components	Coordinates the complete booking flow from seat selection to ticket generation.	Must not permanently store movie or customer data.

MOVIE TICKET BOOKING SYSTEM - UML DIAGRAM



Sequence Diagram - Book a ticket and make Payment

